

# MUHAMMED MAHSHOOK ALI P

## AI ENGINEER | APPLIED AI & LLM APPLICATIONS

Kochi, Kerala, India | +91-7034049672 | mmahshookalip@gmail.com

GitHub: github.com/Mahsh123 | LinkedIn: linkedin.com/in/muhammed-mahshook

### PROFESSIONAL SUMMARY

AI-focused Software Engineer with approximately 1 year of experience building production applications and developing practical AI-powered features for enterprise and learning platforms. Hands-on experience building RAG-based applications, context-aware AI workflows, LLM integrations, speech-enabled systems using STT/TTS, and full-stack software services. Combines Applied AI expertise with strong backend, API, database, cloud, and deployment fundamentals to build production-ready AI systems.

### CORE SKILLS

**Applied AI & LLMs:** Generative AI, LLM Applications, OpenAI API, Gemini API, Prompt Engineering, Context-Aware AI Systems

**RAG & Retrieval:** Retrieval-Augmented Generation (RAG), Document Processing, Embeddings, Vector Search, Semantic Retrieval

**AI Applications:** AI Learning Systems, Conversational AI, Sales Call Simulation, Speech-to-Text (STT), Text-to-Speech (TTS)

**Agentic AI:** Agentic Workflow Concepts, Multi-Step AI Workflows, Tool-Enabled AI Systems

**Software Engineering:** Python, Node.js, TypeScript, JavaScript, React.js, REST APIs, Microservices

**Data & Infrastructure:** PostgreSQL, MySQL, MongoDB, Redis, Docker, Kubernetes, CI/CD, AWS, GCP, Azure

### PROFESSIONAL EXPERIENCE

**Full Stack Engineer** | Aionpixel | Kochi, Kerala | Feb 2026 – Jun 2026

- Built and integrated AI-powered features for enterprise and learning applications, combining full-stack engineering with practical LLM-based capabilities.
- Developed a RAG-based AI learning feature that transformed CBSE textbook content into a retrievable knowledge base, enabling context-aware responses grounded in relevant learning material.
- Built a context-aware AI sales call simulation system using RAG-based context to support realistic, scenario-driven conversations and training workflows.
- Integrated Speech-to-Text (STT) and Text-to-Speech (TTS) capabilities to enable interactive voice-based AI experiences.
- Worked with both open-source and commercial AI models based on feature requirements, evaluating suitable approaches for application quality, capability, and integration needs.
- Developed and maintained full-stack and backend services supporting AI features using modern web technologies, APIs, databases, and enterprise application architecture.
- Designed and integrated AI capabilities into business workflows, focusing on practical automation and production-oriented software systems.
- Contributed to backend architecture, REST APIs, databases, microservices, containerized deployment, and CI/CD practices supporting scalable application delivery.

### AI & DATA SCIENCE EXPERIENCE

**Project Intern – Data Science & AI** | Infosys Springboard | Remote | Nov 2024 – Feb 2025

- Built CareConnect, a voice-based application using React Native, Node.js, MongoDB, and NLP APIs; improved API performance and strengthened testing coverage.

**AI & Cloud Computing Intern** | Edunet Foundation | Remote | Jul 2024 – Aug 2024

- Built AI models on IBM Cloud using Watson AI and gained hands-on exposure to cloud deployment and AI/ML application workflows.

### SELECTED PROJECTS

**CareConnect – Voice-Based Patient System** | React Native, Node.js, TypeScript, MongoDB, NLP APIs

- Developed a voice-enabled application with RESTful backend services and NLP integration.

**Zyppa – Hyperlocal Marketplace & Delivery Platform** | Flutter, REST APIs, Firebase

- Built application workflows for local ordering, delivery coordination, real-time updates, and service interactions.

**TrashEcoMate – Smart Waste Management** | Flutter, Firebase, Python

- Developed an IoT-connected application with real-time data synchronization and a Python-based analytics backend.

**EDUCATION**

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**B.Tech in Computer Science and Engineering** | APJ Abdul Kalam Technological University | 2021 – 2025

**CERTIFICATIONS**

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- Artificial Intelligence Fundamentals – IBM
- Introduction to Generative AI – Google